

Hierarchical Intelligent Operation of Energy Storage Systems in Power Distribution Grids

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Abstract—High penetration of distributed energy storage systems (ESS) offers an unparalleled opportunity to reinforce the distribution grid at the local level against upstream disruptions; however, their mass operation under uncertainty of load and renewable generation is computationally expensive. While deep reinforcement learning (DRL) has been suggested to train operator agents capable of handling uncertainty and high dimensionality of the problem, it falls short when safety and feasibility assurances are required in critical operations. This paper proposes a model for hierarchical coupling of DRL and mathematical optimization for operation of ESS in distribution grids, in order to take advantage of DRL fast response while keeping network constraints in check. In the proposed method, strategic scheduling of distributed ESS units are performed locally by fast DRL-trained agents, while critical grid-wide operations such as fault management and voltage control are performed by an optimization-based central controller. The local controller is trained by Twin Delayed Deep Deterministic Policy Gradient (TD3), whose response time is three orders of magnitude faster than stochastic optimization, while the optimality of solutions are similar in both cases.

Index Terms—Distribution grids, deep reinforcement learning, hierarchical operation, energy storage systems.

I. INTRODUCTION

A. Background and Literature Review

High penetration of small-scale distributed energy resources (DER) including energy storage systems (ESS), has offered an unprecedented opportunity to reinforce the grid at the local level and reduce vulnerability to upstream disruptions [1], [2]. ESS in particular offers operational flexibility and efficiency. As of 2020, more than 20 GWh of ESS has been installed in power grids around the world, in a variety of forms such as batteries, and it is anticipated to grow to 160 GWh by 2030 [3]. Despite the promising growth in penetration of ESS in power grids, the mass operation of these distributed units is quite challenging.

Manuscript received 5 December 2021; revised 1 April 2022 and 3 July 2022; accepted 23 July 2022. Date of publication 16 November 2022; date of current version 22 March 2023. This paper is partly based upon work supported by the U.S. Department of Energy's Office of Energy Efficiency and Renewable Energy (EERE) under the Solar Energy Technologies Office Award Number DE-EE0008775. The views expressed herein do not necessarily represent the views of the U.S. Department of Energy or the United States Government. Paper no. TSTE-01218-2021. (*Corresponding author: Masood Parvania.*)

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Aspects of the technology described in this paper are the subject of a United States patent application.

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TSTE.2022.3222425>.

Digital Object Identifier 10.1109/TSTE.2022.3222425

ESS operation requires strategic response in normal operation, where they provide economic operational flexibility, and prompt response in emergency, where they are used to restore critical loads after outages.

The optimal scheduling of ESS in power distribution grids is extensively studied in the literature. Optimization models has been developed in [4], [5], [6], [7] to schedule ESS in distribution systems for providing services in electricity markets and for countering the uncertainty of renewable energy sources. Multi-level and hierarchical mixed-integer programming has been also used for the operation of ESS units in distribution grids [8]. However, by increasing the number of ESS units and uncertainty sources (such as renewable generation), the dimensionality of the stochastic optimization problems increases, which makes their solution too slow and intractable for real-time operation applications. Various techniques has been suggested to mitigate this curse of dimensionality through dynamic programming [9]. For instance, stochastic dual dynamic programming has been proposed in [10] to capture the temporal dependencies of uncertain variables and to reach a higher computational efficiency. Heuristic methods are also investigated to minimize the operation cost of ESS under uncertainty (see e.g., [11]).

Capabilities of deep reinforcement learning (DRL) in finding fast and near-optimal solutions in stochastic high-dimensional environments has led to its extensive use in operational research, including DER scheduling in power distribution grids [12], [13]. A variety of DRL algorithms with continuous or discrete action domain has been adopted for optimal operation of distributed resources. Using simple actor-critic models [14] or more advanced Asynchronous Advantage Actor-Critic (A3C) models [15] are common approaches to generate continuous control signals for DER. The scalability issue of using DRL for scheduling numerous DER is addressed in [16] by using Deep Deterministic Policy Gradient (DDPG) and breaking down the action space into a sequence of smaller actions. Optimizing ESS charging schedule, in particular, has been a popular application of DRL, as it requires decision-making with long-term considerations. Discrete reinforcement learning (RL) [17] and deep Q-learning [18] are two effective methods investigated for selecting the dispatch level of ESS and optimal energy management of batteries, with the objective of maximum self-sufficiency. Less common RL methods, such as model-free state-action-reward-state-action approach has been also studied in [19] to manage the uncertainty of wind generation using ESS.

Even though DRL can very well handle uncertainty and high-dimensionality, it falls short when feasibility and safety

assurances are required. Although unsafe actions are penalized during the DRL training, there is no guarantee that the output domain of the DRL-trained agent does not overlap the unsafe or infeasible space. This is particularly concerning when scheduling ESS units coincide with critical operations such as fault isolation, load shedding, and voltage control, and taking wrong actions would be quite costly. A viable solution is to divide operational tasks between intelligent and optimization-based algorithms, such that high-dimensional decision are assigned to intelligent and data-driven methods, leaving the optimization method with a smaller problem of taking critical actions, which can be solved quickly and efficiently. As an example, Deep Q-Network (DQN), is utilized in [20] to control voltage regulation devices in slow time scale, while integer optimization is simultaneously used to control fast devices. In [21], the multiaccess edge computing in microgrids is decomposed into a high-level task assignment problem that is solved by combinatorial optimization, and an energy supply planning that is solved by a model-based DRL approach. Data-driven methods have been also used to simulate constraints [22] or to enhance searching efficiency [23] of integer programming in the optimal operation of power systems. Prior efforts for combining DRL and gradient-based optimization for grid operation are either focused on specific operational tasks, e.g., [21], or on improving the optimization performance, as in [22], [23]. A comprehensive DRL-aided operational framework that can schedule numerous DER units under uncertainty, while making safe decisions for critical tasks such as fault management, is yet to be developed. To the best of our knowledge, no prior effort has been made to combine data-driven and optimization based methods for fast and yet reliable operation of ESS in distribution grids.

B. Contribution and Paper Structure

Addressing the gap in the literature, this paper proposes a hierarchical intelligent operation model for distribution grids operation, in which scheduling the ESS operation and meeting local demand is performed locally, while a central controller is responsible for critical grid-wide operations such as voltage control and fault management. In the proposed model, shown in Fig. 1, first, the power distribution grid is divided into multiple zones that are separated by inter-zonal tie-switches. Each zone contains a local controller (LC) that decides on active power dispatch of ESS within the zone, based on the predicted local load, energy price, and renewable generation. DRL is implemented using Twin Delayed DDPG (TD3) algorithm that is a robust version of DDPG against training perturbations, and is used to train independent LCs for each zone. A central controller (CC), located at the substation level, receives local setpoints from LCs and outage data from data acquisition modules and executes a power flow and network reconfiguration, where each zone is seen as a single node with a net power demand. The CC decides on switching operations and determines required reactive power on each zone to maintain all voltages within permissible limits. Finally, LCs receive the reactive power setpoint from the CC and dispatch the ESS within their zones accordingly.

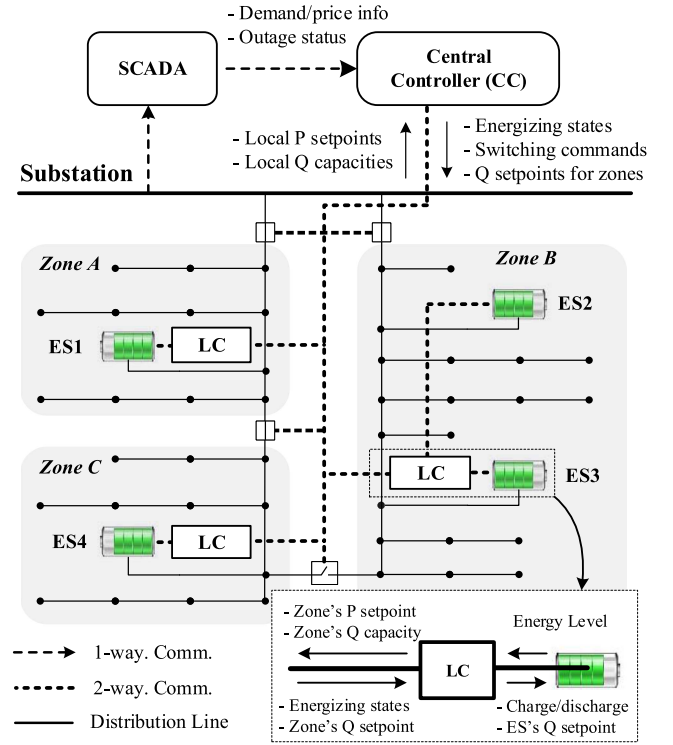


Fig. 1. Proposed model for hierarchical operation of energy storage systems.

In the proposed hierarchical model, the division of tasks between DRL and mathematical optimization results in a fast response for the overall model that reduces the real-time operation scheduling time and enables the scalability of coordinating the operation of large number of ESS in distribution grids. Also, the use of DRL-based local controllers in the proposed model is a more practical solution, compared to similar methods such as distributed optimization. The use of mathematical optimization for local operation in each zone requires a processing unit to regularly run the optimization model. That would be challenging because distribution zones do not have a separate substation, and processing power is not readily available everywhere. The proposed DRL-based local controllers, that are trained offline, are simply neural networks with fixed weights and bias values. They can be implemented on field-programmable gate arrays without the need for a separate processing unit.

The rest of the paper is organized as follows: the structure of the hierarchical intelligent operation model, and working mechanisms of LC and CC are presented in Section II. Section III presents the training procedure of LC using Twin Delayed DDPG. The results of testing the proposed operation model on the 123-bus test distribution grid is presented in Section IV, and Section V concludes the paper.

II. HIERARCHICAL OPERATION MODEL FOR DISTRIBUTED ENERGY RESOURCES

The proposed hierarchical operation model consists of a central controller (CC) in the substation, multiple local controllers (LC) in various distribution zones, and a designated flow of

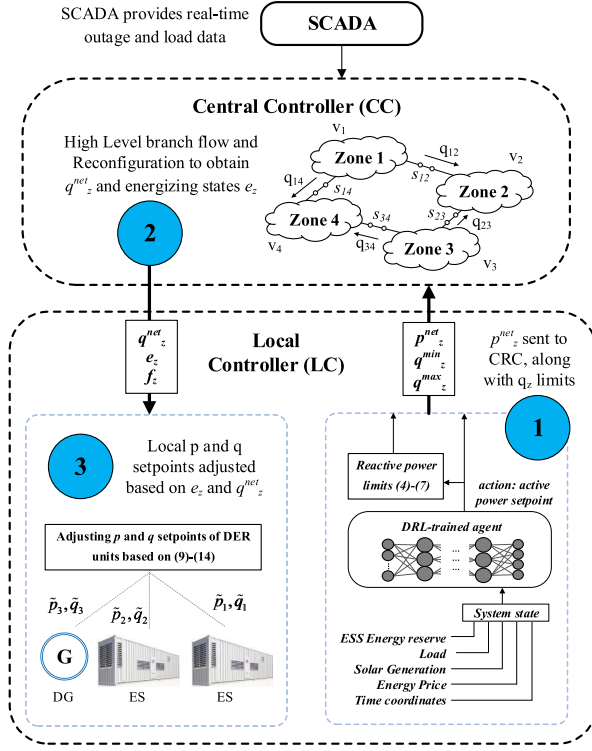


Fig. 2. The three stages of implementing the proposed hierarchical operation model. In each time-step, these stages determine the active and reactive power setpoints of each ESS unit.

communication and commands between them, as shown in Fig. 1. Within each zone, the active power dispatch of ESS, as well as upper and lower bounds on reactive power generation is determined by the LC. In the grid level, the CC solves a power flow problem for the whole system, where each zone is seen as a node with a net power demand, and determines the reactive power required at each node (i.e., zone) to maintain the voltage within limits. Solving the distribution-level problem, where each distribution zone is treated as a single node, requires that voltage deviation within each zone be limited by δ , and then the distribution-level operation problem be solved with voltage limits tightened by δ . In other words, the distribution grid is divided to zones such that:

$$|V_{it} - V_{jt}| < \delta, \forall t, \forall i, j \in \mathcal{I}_z, \forall z \in \mathcal{Z}, \quad (1)$$

where $\mathcal{Z}$ is the set of all zones, t is the time index, and $i, j \in \mathcal{I}_z$ represent nodes in zone $z \in \mathcal{Z}$. Once the CC determines the reactive power setpoint for each zone, it sends them back to LCs that use it to schedule the reactive power of all inverter-based DERs (ESS and DG units) within their zones. The CC also operates tie-switches to isolate zones that contain fault and restore power to healthy interrupted zones. In zones that are isolated by the CC due to existence of a fault, all DERs within the zone will be shut down. Fig. 2 illustrates the division of operational tasks between CC and LCs, and the hierarchical sequence of decision making in each time step. In the following, the detailed operation of both LC and CC are explained in three steps: generating local setpoints, generating global setpoints, and revising local setpoints.

A. Generating Local Setpoints

The LC within a distribution zone controls the energy level of ESS units in the same zone by issuing charging or discharging commands, resulting in the output power of $-p_{it}^c$ or p_{it}^d for the ESS on node i at time t . The net active power output of the ESS is shown by $p_{it}^{ESS} = p_{it}^d - p_{it}^c$. The objective of the LC operation within each zone is to optimize ESS charging/discharging power such that the operation cost is minimized, while respecting ESS operational constraints. Therefore, the following optimization problem should be solved by each LC:

$$P1: \min_{p_{it}^d, p_{it}^c} \sum_{t=1}^T \lambda_t^E \left(\sum_{i \in \mathcal{I}_z} p_{it}^L - \sum_{i \in \mathcal{G}_z} p_{it}^g - \sum_{i \in \mathcal{E}_z} p_{it}^{ESS} \right), \quad (2a)$$

$$\text{s.t. } E_{i,t+1} = E_{it} + \eta^c p_{it}^c - \frac{1}{\eta^d} p_{it}^d, \forall i \in \mathcal{E}, \forall t, \quad (2b)$$

$$0 \leq p_{it}^d \leq \bar{p}_{it}^d, \quad 0 \leq p_{it}^c \leq \bar{p}_{it}^c, \forall i \in \mathcal{E}, \forall t, \quad (2c)$$

$$p_{it}^{ESS} = p_{it}^d - p_{it}^c, \forall i \in \mathcal{E}, \forall t, \quad (2d)$$

$$E_{it} - \bar{E}_i \leq p_{it}^{ESS} \leq E_{it} - \underline{E}_i, \forall i \in \mathcal{E}, \forall t, \quad (2e)$$

where λ_t^E is electricity price, p_{it}^L is active power demand, and p_{it}^g is non-dispatchable generation (e.g., solar generation) on node i at time t . Also, $\mathcal{E}_z, \mathcal{G}_z \subset \mathcal{I}_z$ are the set of nodes with ESS and DG units, respectively. The evolution of ESS energy level (E_{it}) by charging and discharging actions is defined in (2b), where charging and discharging power losses are accounted for by efficiency factors η^c and η^d , respectively. Further, the charging and discharging rates are limited to $\bar{p}_{it}^c$ and $\bar{p}_{it}^d$ in (2c), and ESS net active power output is limited by its energy reserve in (2e).

Solving P1 requires accurate information on load and generation in future hours, which depend on multiple uncertain factors such as renewable generation, weather events and customer behavior. It also requires knowledge of future changes in the energy price. In this paper, instead of solving P1 using stochastic and probabilistic methods, we propose using DRL to capture uncertainties through observing real data samples, and ultimately training a controller that decides on ESS scheduling, as explained in detail in Section III. Once P1 is solved at the zone level, the net active power demand of the zone p_{zt}^{net} , as given by (3), is sent to the CC:

$$p_{zt}^{net} = \sum_{i \in \mathcal{I}_z} p_{it}^L - \sum_{i \in \mathcal{E}_z} p_{it}^{ESS} - \sum_{i \in \mathcal{G}_z} p_{it}^g \quad (3)$$

The other two signals that are generated at the zone level, are upper and lower limits of total reactive power capacity within the zone. Typically, ESS and renewable generation units, are connected to the grid via inverters with limited power capacity. The reactive power generated by the inverter installed for ESS on bus i is constrained by $|q_{it}^{ESS}| \leq \sqrt{(\bar{s}_i)^2 - (p_{it}^{ESS})^2}$ where $\bar{s}_i$ is the apparent power capacity of the i -th inverter. Traditionally, inverters are designed as $\bar{s}_i = \bar{p}_i^{ESS}$, where $\bar{p}_i^{ESS}$ is the active power rating of the unit. However, when maximum output is reached, i.e., $p_{it}^{ESS} = \bar{p}_i^{ESS}$, no reactive power can be generated. To address this, oversized inverters' nameplate capacity has been

advocated such that $\bar{s}_i > \bar{p}_i^{ESS}$ [24]. For instance, by choosing $\bar{s}_i = 1.08\bar{p}_i^{ESS}$ and limiting q_{it}^{ESS} to $\sqrt{(\bar{s}_i)^2 - (\bar{p}_i^{ESS})^2}$ instead of $\sqrt{(\bar{s}_i)^2 - (p_{it}^{ESS})^2}$, the reactive power compensation provided by inverter i is $|q_{it}^{ESS}| < 0.4\bar{p}_i^{ESS}$, regardless of the instantaneous ESS output. As such, the reactive power generated by inverter i is constrained by:

$$|q_{it}^{ESS}| \leq \bar{q}_{it}^{ESS} = \sqrt{(\bar{s}_i)^2 - (\max\{-\bar{p}_i^c, \bar{p}_i^d\})^2}, \quad \forall i \in \mathcal{E}, \forall t, \quad (4)$$

The same equation holds true for inverter-based generation:

$$|q_{it}^g| \leq \bar{q}_{it}^g = \sqrt{(\bar{s}_i)^2 - (\bar{p}_i^g)^2}, \quad \forall i \in \mathcal{G}, \forall t. \quad (5)$$

Therefore, the limits on net reactive power consumption in each zone are given by:

$$q_{zt}^{\min} = \sum_{i \in \mathcal{I}_z} q_{it}^L - \sum_{i \in \mathcal{E}_z} \bar{q}_{it}^{ESS} - \sum_{i \in \mathcal{G}_z} \bar{q}_{it}^g, \quad \forall z, t, \quad (6)$$

$$q_{zt}^{\max} = \sum_{i \in \mathcal{I}_z} q_{it}^L - \sum_{i \in \mathcal{E}_z} \bar{q}_{it}^{ESS} - \sum_{i \in \mathcal{G}_z} \bar{q}_{it}^g, \quad \forall z, t. \quad (7)$$

Once local setpoints are generated, the LC sends three signals to the CC: p_{zt}^{net} , $q_{zt}^{\min}$, $q_{zt}^{\max}$.

B. Generating Global Setpoints

At the grid level, the CC receives preliminary decisions on the net active power generation of zones, as decided by LCs within each zone, and runs a distribution-level branch flow optimization model to find the required reactive power in each zone to maintain voltage levels between desired limits. The CC treats each zone as a single node with a connected lump load equal to the zone's total net power demand. It also controls the tie-switches between zones, and may change them if fault isolation and load restoration is required. To that end, the CC solves the following mixed-integer linear model, in which the number of variables is only proportional to the number of zones. In this model, the net active power of each zone p_{zt}^{net} , which is determined by the local controllers, is an input parameter, while control variables include net reactive power on each node (zone) and state of tie-switches. Since this optimization model is solved independently at each time interval, the subscript t is eliminated from variables. Note that in the following formulation, the constraints are true for all subscripts, unless otherwise specified.

$$\min_{e_z, q_z^{net}} \sum_{z \in \mathcal{Z}} (1 - e_z) p_z^{net} + \sum_{zz' \in S_w} \frac{1}{M} (1 - s_{zz'}) \quad (8a)$$

s.t.

$$p_1^{net} = p^{\text{grid}}, \quad q_1^{net} = q^{\text{grid}}, \quad (8b)$$

$$\sum_{z'|(z,z') \in \mathcal{L}} p_{zz'} - p_{z'z} + r_{zz'}(p_{z'z}^2 + q_{z'z}^2) \leq -e_z p_z^{net}, \quad (8c)$$

$$\sum_{z'|(z,z') \in \mathcal{L}} q_{zz'} - q_{z'z} + x_{zz'}(p_{z'z}^2 + q_{z'z}^2) \leq -q_z^{net}, \quad (8d)$$

$$e_z q_z^{\min} \leq q_z^{net} \leq e_z q_z^{\max}, \quad (8e)$$

TABLE I
ZONE CONDITION BASED ON EXISTING FAULTS AND ENERGIZING STATE

f_z	e_z	Description	Zone State
0	0	no fault, de-energized	Islanded, $z \in Z^R$
0	1	no fault, energized	Normal $z \in Z^N$
1	0	fault, de-energized	Isolation, $z \in Z^F$
1	1	fault, energized	Energized fault, avoided by (8j)

$$0 \leq (p_{zz'})^2 + (q_{zz'})^2 \leq s_{zz'} \cdot (\bar{S}_{zz'})^2, \quad (8f)$$

$$v_{z'} \leq v_z - 2(r_{zz'} p_{zz'} + x_{zz'} q_{zz'}) + M(1 - s_{zz'}), \quad (8g)$$

$$v_{z'} \geq v_z - 2(r_{zz'} p_{zz'} + x_{zz'} q_{zz'}) - M(1 - s_{zz'}), \quad (8h)$$

$$\underline{v} + \delta \leq v_z \leq \bar{v} - \delta, \quad (8i)$$

$$e_z \leq 1 - f_z, \quad (8j)$$

$$e_{z'} - M(1 - s_{zz'}) \leq e_z \leq e_{z'} + M(1 - s_{zz'}), \quad (8k)$$

$$e_z \leq \sum_{zz' \in L} s_{zz'}, \quad (8l)$$

$$\sum_{zz' \in L} s_{zz'} < |L|, \quad \forall L \in \text{loops}. \quad (8m)$$

The objective function (8a) is to minimize the loss of load and the number of switching, where e_z and $s_{zz'}$ are binary variable that respectively represent the energizing condition of each zone and the state of tie-switch between zones z, z' . In normal conditions, in which no loss of load is expected, the model may change the state of switches and adjust reactive power flow to keep all voltages within their limits and also to meet other grid constraints. When a fault occurs in one of the zones, the model minimizes load interruption as the main objective by changing the state of switches, while still meeting network constraints. The quadratically-constrained branch flow equations are formulated in (8b)–(8i). In (8d), variable q_z^{net} is the net reactive power requirement on zone z to avoid voltage violation, and is limited by locally generated setpoints in (8e). The voltage limit on each zone is restricted by (8i), which tightens the normal bounds by δ to take into account small voltage variations within each zone. (8j)–(8l) represent switching constraints for fault isolation and service restoration. The fault isolation is ensured in (8j), where f_z is a binary parameter indicating the existence of fault in z . The network connectivity and integrity of the energy flow is checked by (8k). For example, if the tie-switch between z, z' is closed ($s_{zz'} = 1$), then the energizing states of the two zones should be the same, otherwise their energizing states are independent. Further, (8l) ensures de-energizing of zones with no closed ti-switches to other zones. The radiality of the network given the existing *loops* is checked in (8m).

When the CC solves the problem above, it sends back three signals to each zone: $f_{zt}, e_{zt}, q_{zt}^{net}$. The combination of energizing state and fault state signals, e_{zt}, f_{zt} , will determine if the zone is in the normal, faulted, or islanded condition, as shown in Table I. For instance, if there is no fault in the zone but it is de-energized ($f_z = 0, e_z = 0$), it means the zone is healthy but islanded and is prepared for restoration. Those signals will be used to revise the local setpoints.

C. Revising Local Setpoints

The LCs then recalculate the setpoints for local DERs based on the signals received from the CC. The recalculated setpoints, denoted by $(\hat{\cdot})$, are obtained as described next.

1) *Faulted Zones*: If there is a fault in zone z ($f_z = 1$), all local generation units must be shut down. Therefore:

$$\{\hat{p}_{it}^g, \hat{q}_{it}^g, \hat{p}_{it}^{ESS}, \hat{q}_{it}^{ESS}\} = (1 - f_z) \cdot \{p_{it}^g, q_{it}^g, p_{it}^{ESS}, q_{it}^{ESS}\}, \quad \forall z \in Z^F, \forall i \in \mathcal{I}_z. \quad (9)$$

2) *Normal and Islanded Zones*: In zones with normal condition, where $f_z = 0, e_z = 1$, there will be no change in the original setpoints for active power generation.

$$\{\hat{p}_{it}^g, \hat{p}_{it}^{ESS}\} = \{p_{it}^g, p_{it}^{ESS}\}, \forall z \in Z^N, \forall i \in \mathcal{I}_z. \quad (10)$$

If zone z is islanded ($f_z = 0, e_z = 0$), the LC commands all ESS in the zone to discharge at full capacity until the zone is restored or ESS runs out of charge. However, the discharge amount should not exceed the net demand so the power balance is maintained within the zone. The required active power in the zone are divided among available ESS, based on their capacities. The revised setpoints in islanded zones ($\forall z \in Z^R, \forall i \in \mathcal{I}_z$) are given by:

$$\hat{p}_{it}^g = p_{it}^g, \hat{p}_{it}^{ESS} = \bar{p}_{zt}^{ESS} \times \frac{\bar{p}_{it}^{ESS}}{\sum_{i' \in \mathcal{E}_z} \bar{p}_{i't}^{ESS}}, \quad (11)$$

$$\bar{p}_{zt}^{ESS} = \min \left(p_{zt}^L - p_{zt}^g, \sum_{i \in \mathcal{E}_z} E_{it} - \underline{E}_{it} \right). \quad (12)$$

In both normal and islanded conditions, the setpoints for reactive power generation of DER units are obtained by dividing the reactive power setpoint, received from CC, among DER units of the zone based on their capacity. The reactive power dispatch of DER units in both normal and islanded zones ($\forall z \in Z^N \cup Z^R, \forall i \in \mathcal{I}_z$) are given by:

$$\hat{q}_{it}^{ESS} = (-q_z^{net} + q_z^L) \times \frac{\bar{q}_{it}^{ESS}}{\sum_{i' \in \mathcal{E}_z} \bar{q}_{i't}^{ESS} + \sum_{i' \in \mathcal{G}_z} \bar{q}_{i't}^g}, \quad (13)$$

$$\hat{q}_{it}^g = (-q_z^{net} + q_z^L) \times \frac{\bar{q}_{it}^g}{\sum_{i' \in \mathcal{E}_z} \bar{q}_{i't}^{ESS} + \sum_{i' \in \mathcal{G}_z} \bar{q}_{i't}^g}. \quad (14)$$

The steps by which the hierarchical operation is implemented are shown in Fig. 2, where the first step is implemented by training an agent using deep reinforcement learning. Details of training this agent are explained in the next section.

III. REINFORCEMENT LEARNING FOR GENERATING LOCAL SETPOINTS

In the hierarchical resilient operation, the local setpoints for active power generation are produced using LC which is trained by DRL. Applying DRL requires the problem to be presented as a Markov Decision Process (MDP), which can be characterized by the following parameters:

- *State*: denoted by set $\mathcal{S}$ and includes system observations, based on which operational decisions are made. Each system state $s_t \in \mathcal{S}$ includes a set of parameters in time

slot t that affect the choice of action, and is defined as the following tuple:

$$s_t = (d_t^y, d_t^w, h_t, \mathbf{E}_t, p_t^L, \tilde{p}_t^L, p_t^g, \tilde{p}_t^g, \lambda_t, \tilde{\lambda}_t), \quad (15)$$

where d_t^y, d_t^w and h_t represent day of the year, day of the week, and the hour of the day, respectively, and are important since they affect the variation of load and energy price profiles. Also, $\mathbf{E}_t$ is energy level vector of all ESS units within the zone. The notation $(\tilde{\cdot})$ denotes the average look-ahead values. Since both load and energy price profiles follow daily cycles, 24 hours is taken as the look-ahead horizon.

- *Actions*: denoted by set $\mathcal{A}$, where each action $a_t \in \mathcal{A}$ is the net output power of ESS in the zone and is given by $a_t = \mathbf{p}_t^{ESS}$, where $\mathbf{p}_t^{ESS}$ is the vector of active power dispatch of all ESS within the zone.
- *Cost function*: defined as $\mathcal{C} : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{R}$, is the cost of taking control action a_t at state s_t . One may simply take the operation cost as the cost function, but that includes the cost of load consumption as well, and therefore is not a proper metric to evaluate how well the actions are taken. Instead, we only take the cost of ESS charging/discharging as the cost function. In addition, a regularization term is embedded in the cost function to avoid unnecessary large actions, resulting in:

$$\mathcal{C}_t = \mathcal{C}(s_t, a_t) = \sum_{i \in \mathcal{E}_z} \lambda_t p_{it}^{ESS} + m \|\mathbf{p}_{it}^{ESS}\|. \quad (16)$$

- *Transition function*: defined as $\mathcal{P} : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow [0, 1]$, and determines the probability of arriving at state s_{t+1} by taking action a_t in state s_t . This probability that is governed by the prediction error of load, generation, energy price, and other uncertainty sources in the system, is hard to obtain. DRL does not model this transition function, but implicitly learns it through observing many transitions.

Using the MDP framework above, we aim to minimize the total discounted cost of ESS operation and solve the following problem P2 instead of P1 (2a):

$$\begin{aligned} \text{P2: } \min_{\pi} \quad & \mathbb{E}_{s, a \sim \pi} \left[\sum_{t=1}^{\infty} \gamma^t [\mathcal{C}(s_t, a_t)] \middle| s_0 = s \right], \\ \text{s.t. } & (2b)-(2e), \end{aligned} \quad (17)$$

where the expectation is over transition function $\mathcal{P}$, $\gamma \in [0, 1]$ is a discount factor that signifies the importance of future costs compared to current cost, and $\pi(s) = a$ is the action policy. As already mentioned, in real-world power systems, the prediction error of the demand/generation does not follow a certain distribution, leading to unknown transition function $\mathcal{P}$. Therefore, problem P2 cannot be solved explicitly, however, a near-optimal solution may be found using DRL that finds highly rewarded decision trajectories by observing many state transitions and learning the transition function.

A. DRL Implementation by Twin Delayed DDPG

In this work, TD3 algorithm, which is a robust variation of DDPG, is used to implement DRL and identify optimal policies in a continuous action space. In the proposed algorithm, the long-term value of taking action $a_t \in \mathcal{A}_t$ in each system state is estimated by a Q-value, $Q(s_t, a_t)$, which will be used to find the most valuable actions. The Bellman equation gives a recursive formula for obtaining the Q-value of action a_t at state s_t , based on the immediate reward and the Q-values in the next step.

$$Q(s_t, a_t) = -C_t + \gamma \mathbb{E} \left[\max_{a_{t+1} \in \mathcal{A}_{t+1}} Q(s_{t+1}, a_{t+1}) \right], \quad (18)$$

In the DDPG method, two independent deep neural networks are trained to recursively find Q-values; an actor network $\mu(s_t; \theta^\mu)$ to estimate the most rewarded action in state s_t , and a critic network $Q(s_t, a_t; \theta^Q)$ to estimate the Q-values of taken actions. The weight vectors of the actor and critic networks are marked by θ^μ, θ^Q , respectively. Given properly trained networks, the Q-value is estimated as follows:

$$Q(s_t, a_t) \approx -C_t + \gamma \mathbb{E} [Q(s_{t+1}, \mu(s_{t+1}; \theta^\mu); \theta^Q)]. \quad (19)$$

These networks are trained in parallel, by observing system states, taking actions by the actor network, and observing the reward and the new state, all of which provide a training sample for both actor and critic networks. More specifically, the critic network is trained on samples of s_t, a_t as input, where $a_t = \mu(s_t; \theta^\mu)$, and the reward r_t as output. The actor network takes s_t as input and the $\max_a Q(s_t, a_t; \theta^Q)$ as output. For better exploration of the action space and to avoid overfitting, a random noise $\mathcal{N}_t$ is also added to the decision-making mechanism, resulting in $a_t = \mu(s_t; \theta^\mu) + \mathcal{N}_t$. In order to meet the constraints of ESS operation, (2b)–(2e), actions a_t are modified as:

$$a_t = \max(-\bar{p}_t^c, \min(\bar{p}_t^d, a_t)), \quad (20)$$

$$a_t = \max(E_{i,t} - \bar{E}_i, \min(E_{i,t} - \underline{E}_i, a_t)). \quad (21)$$

It has been shown that for stabilizing the training process, it is useful to have target networks, which are structural copies of the original actor and critic networks. The target actor and critic networks, whose weight vectors are shown by $\theta^{\mu'}, \theta^{Q'}$ respectively, are not trained directly, but their weights are gradually updated by the original networks. The target networks act as relatively more stable references to find the loss function. Also, instead of calculating the total expectation as in (19), it is more computationally efficient to update the weights using samples from a batch of recent observed transitions U , where $|U| = N$. Thus, the critic network is trained by minimizing the following loss function:

$$L(\theta^Q) = \frac{1}{N} \sum_i \left[(Q(s_t, \mu(s_t; \theta^\mu); \theta^Q) - y_i)^2 \right], \quad (22a)$$

$$y_i = -C_t + Q(s_{t+1}, \mu(s_{t+1}; \theta^{\mu'}); \theta^{Q'}),$$

$$i := (s_t, a_t, r_t, s_{t+1}) \sim U \quad (22b)$$

and updating the weights as $\theta^Q = \theta^Q + \eta^Q \nabla_{\theta^Q} L(\theta^Q)$ in each training step. The actor network is trained similarly, but the

TABLE II
SPECIFICATIONS OF ESS AND PV UNITS IN THE TEST SYSTEM

Zone	1	2	3	4	5	6
ESS capacity (kWh)	1000	1000	2000	2000	1000	2000
ESS power rate (kW)	250	250	500	500	250	500
PV power rate (kW)	130	125	250	300	125	250

gradient of its loss is obtained by the gradient chain rule:

$$\begin{aligned} \nabla_{\theta^\mu} L(\theta^\mu) \\ = \frac{1}{N} \sum_i \nabla_a Q(s_t, \mu(s_t; \theta^\mu); \theta^Q) \cdot \nabla_{\theta^\mu} \mu(s_t; \theta^\mu). \end{aligned} \quad (23)$$

Although DDPG method is usually efficient in finding highly rewarded decision trajectories, it suffers from overestimation of the Q-function, which could derail the learning process by exploiting the estimation error. As a result a more robust variation, called Twin Delayed DDPG (TD3), is proposed in [25] that considerably reduces the overestimation. In this method, two separate target networks are trained in parallel for the critic network, which results into two separate loss functions, $L(\theta_1^Q), L(\theta_2^Q)$. In each iteration, the minimum estimation of the two networks is used to calculate y_i , replacing (22b) with:

$$y_i = -C_t + \min_{i \in \{1,2\}} Q(s_{t+1}, \mu(s_{t+1}; \theta^{\mu'}); \theta_i^{Q'}). \quad (24)$$

Further, the training steps are delayed such that the policy is updated for every two Q-function updates. Finally, a soft update is utilized to gradually update the target networks based on the original networks using the update coefficient τ :

$$\theta_i^{Q'} = \tau \theta_i^Q + (1 - \tau) \theta_i^{Q'} \quad i \in \{1, 2\}, \quad (25a)$$

$$\theta^{\mu'} = \tau \theta^\mu + (1 - \tau) \theta^{\mu'}. \quad (25b)$$

The training process goes on for as many episodes as needed to converge the Q-values. Once the network is trained and the loss function converges to below a certain value, the trained model is detached from the training setup and used directly to schedule ESS units within each zone.

IV. CASE STUDY

The proposed hierarchical intelligent operation model is tested on a modified IEEE 123-bus test distribution grid, consisting of 6 zones interconnected via tie-switches, where each zone contains an ESS and a photovoltaic (PV) unit, as shown in Fig. 3. Charging and discharging efficiency of all ESS is set to 0.9. Power rates and energy capacity of ESS and PV units are chosen proportional to the total installed load in their zones, and are presented in Table II. The reactive power capacity of all inverter-based units are obtained using (4),(5) assuming that $|q_{it}^{ESS}| \leq 0.4 \bar{p}_i^{ESS}$ and $|q_{it}^g| \leq 0.4 \bar{p}_i^g$.

For the real-time and 24-hour look-ahead profile of demand and electricity price, the NYISO 2017 hourly data [26] is obtained for six of its operational zones, and are mapped to zones 1 to 6 of the 123-bus test system by normalizing them to the total load installed in each zone. National Solar Radiation DataBase

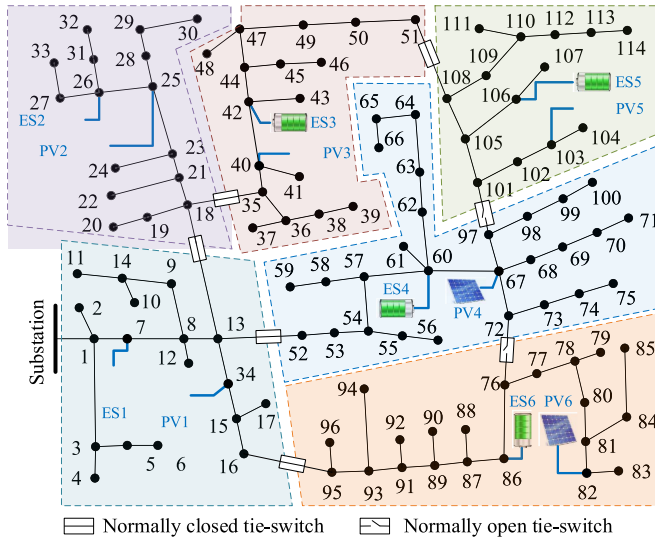


Fig. 3. Modified IEEE 123-bus test distribution grid, divided into 6 zones with independent ESS control. For more clarity, the communication links, LCs and CC are not shown.

(NSRDB) [27] is used to extract hourly Global Horizontal Irradiation (GHI) index in 2017 in Manhattan, NY, by which a PV generation profile is created and normalized for each zone based on its PV power rate given in Table II. All aforementioned data are real-world observations without any probabilistic modeling. However, the predicted solar irradiation is modeled by a normal distribution around the actual values, with $\mu = \text{GHI}$, $\sigma = 5$. The resulting database for load and renewable generation is available in [28].

The MDP model is designed using OpenAI Gym toolkit by specifying the state and action spaces, as well as episode reset and step through functions. The Twin Delayed DDPG (TD3) framework is implemented using Keras-rl in Python, where both actor and critic are feed-forward neural networks with three hidden layers with 50, 100 and 50 neurons and a Relu activation function after each layer. The same structure is adopted for the target networks. The hyper-parameters of TD3 are as follows: Adam optimizer performs stochastic gradient with learning rate of $\alpha = 0.0005$, and the discount factor is $\gamma = 0.99$. The rate by which the target network is updated is $\tau = 0.001$. Also, the Ornstein-Uhlenbeck random process is added as noise $\mathcal{N}$ to the actions taken by the actor network with $\beta = 0.2$ and $\sigma = 0.2$ for better exploration of the action space. It was also observed that normalizing the state parameters, such that all parameters are of the same order, improves the final result. As such, the net load and electricity price data samples are normalized by the maximum installed load and average annual price in each zone, respectively. separate LCs are trained in each zone using TD3 method, and for each method the best trained agent from 5 different runs is selected. Fig. 4 shows the convergence trend of the reward during the training process of the LC in zone 1.

For testing, the real-time and look-ahead data of the same network from 2018, which are unseen during the training process, is utilized. In order to test the performance at different times

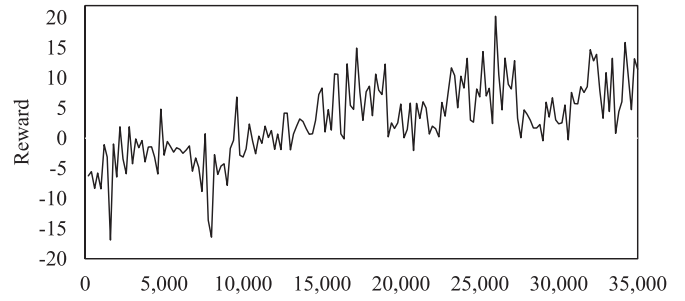


Fig. 4. Improvement of reward function by increasing training steps, during training of the local controller in Zone 1, using TD3 algorithm.

TABLE III
AVERAGE DAILY OPERATION COST OBTAINED BY APPLYING DIFFERENT MODELS ON 60 DAYS OF TEST DATA

Zone	Deterministic	Stochastic	Hierarchical (DDPG)	Hierarchical (TD3)	Perfect knowledge
1	1,340	1,178	1,227	1,260	1,043
2	1,236	1,188	1,139	1,039	877
3	2,413	2,277	2,308	2,225	2,003
4	2,952	2,857	2,915	2,733	2,550
5	887	896	811	797	629
6	2,786	2,791	2,797	2,637	2,309
Total	11,615	11,188	11,197	10,691	9,410

of the year, the hierarchical model is tested on 60 days of data, consisting of four 15-day intervals, each in a different season in 2018. The LCs are trained using DDPG with and without the TD3 structure and the resulting hierarchical model (including LCs and CC) is tested against the test data, the results of which are listed in Table III for average daily operation cost, and are compared to deterministic and stochastic optimization models. For the deterministic model, in each time step, problem P1 (2a) is solved for a 24 h horizon, considering predicted values for the net load and energy price as deterministic.

In the stochastic optimization model, P1 is solved using model predictive control that solves the problem for a sliding window of future hours and for a limited number of scenarios that are generated based on predicted values and a probability distribution for the prediction error. While the results yield an operational solution for the whole window, only the solution of the current time step is used, and in the next time step a new problem is solved. The prediction error for net load and energy price is fitted to a Gaussian Distribution where μ is equal to the predicted values and σ values are equal to 5.5% and 18% of the predicted values of the net load and energy price, respectively. The parameters of the Gaussian Distribution are determined based on the available historical data. The results and computation times reported here for the stochastic optimization model reflect the online decision making using the model predictive control method. The stochastic optimization is first run for five randomly generated scenarios and for a horizon length of 10 time steps. In Table III and Fig. 5, the proposed hierarchical method with DDPG-trained LCs has a somewhat similar performance to the stochastic model, while replacing DDPG with TD3 further reduces the operation cost by average 4.44%. The results of

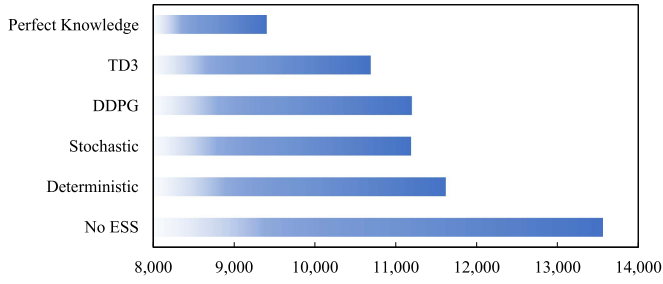


Fig. 5. Comparison of average daily operation cost for applying different operation models on the test data.

TABLE IV
COMPARING THE PERFORMANCE OF DRL-TRAINED MODEL AND STOCHASTIC OPTIMIZATION IN OPERATION OF ESS UNITS WITH CONTINUOUS ACTION SPACE

	Num. of scenarios	Window length	Runtime (sec)	Operation costs (\$)
Stochastic Optimization	1	10	0.78	11,615
		20	1.63	11,509
	5	10	2.75	11,188
		20	5.50	10,948
	10	10	5.44	11,109
		20	11.29	10,747
	20	10	11.66	10,652
		20	26.35	10,298
The proposed hierarchical model (CC + LC)			0.16 (0.15 + 0.01)	10,691

a perfect knowledge case, in which an omniscient operator exactly knows the future values and decides accordingly, is also shown in Table III and Fig. 5 for comparison, and shows the hierarchical model with TD3-trained LCs has performed close to perfection.

In stochastic optimization, increasing the number of scenarios can enhance the optimality of decisions in the uncertain environment, at the expense of higher computation time that increases exponentially with the number of considered scenarios. Table IV shows a comparison between stochastic optimization, run with different number of scenarios and horizon lengths, with the proposed hierarchical model that uses TD3 algorithm to train LCs. The stochastic optimization yields lower operation costs (average value over 60 days of test data) only when the number of scenarios reaches 20. However, the computation time of the stochastic models is at least three orders of magnitude longer than the hierarchical model, which makes the proposed model a more suitable alternative for real-time operation applications where prompt signals are required for voltage and frequency control or switching operations. Note that in the hierarchical model, LCs run in parallel, and therefore their computation time overlaps. The total runtime of the hierarchical model consists of fast response time of DRL-trained LCs (0.01 sec), plus the runtime of CC that solves a small-scale real-time optimization (0.15 sec) over zones. The difference in response time is more evident when integer variables are present. In distribution

TABLE V
COMPARING THE PERFORMANCE OF DRL-TRAINED MODEL AND STOCHASTIC OPTIMIZATION IN OPERATION OF ESS UNITS WITH DISCRETE ACTION SPACE

	Num. of scenarios	Window length	Runtime (sec)	Operation costs (\$)
Stochastic Optimization	1	10	1.87	13,546
		20	3.82	13,385
	5	10	16.09	12,632
		20	58.50	12,934
	10	10	49.18	12,507
		20	114.82	12,153
	20	10	138.41	11,839
		20	411.55	11,559
The proposed hierarchical model (CC + LC)			0.16 (0.15 + 0.01)	11,798

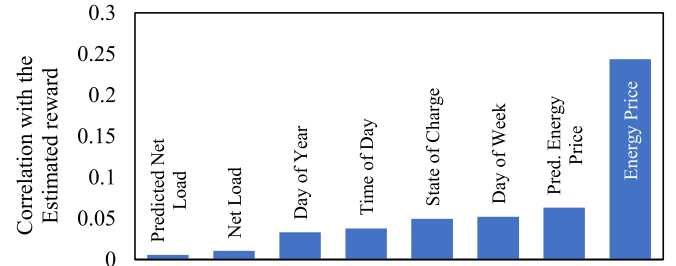


Fig. 6. Correlation between reward value and state parameters, estimated by analyzing the behavior of the trained DRL agent in Zone 1.

grids, integer decision variables are common and represent binary/integer generation setpoints, switching of capacitor banks, and tap changer levels of transformers, among others. To highlight the impact of integer variables on both optimization model and the proposed hierarchical model, we conducted another study where ESS units take integer setpoints, i.e., charging at full capacity, charging at half capacity, idle, discharging at half capacity, and discharging at full capacity. Also, PV units take binary on/off setpoints. The results are presented in Table V and indicate that while integer variables considerably increase the operation time and computational burden of stochastic optimization, they have almost no impact on the response time of the hierarchical model, as the local decision-making mechanism of DRL-trained LCs is not gradient-based.

The superior performance of the DRL-trained agents is due to subtle correlations between selected actions and state parameters, which are learned by the DRL-trained model through the training process. In order to evaluate the correlation between performance and those parameters, 10 k states are randomly generated and fed to the developed DRL-trained agent, and the correlations between each state parameter and the estimated reward is calculated and shown in Fig. 6. As expected, the estimated reward is highly impacted by the real-time and predicted energy price. It also shows that day of the week is a somewhat influential factor in reward estimation and decision-making, which means the DLR-trained agent has learned different load

TABLE VI
THE SEQUENCE OF EVENTS IN A TEST SCENARIO AND CONTROL RESPONSES BY THE PROPOSED HIERARCHICAL OPERATION MODEL

hour	Prelim. power dispatch	Event	Switching	Confirmed power dispatch	Computation time
9	$\mathbf{p}^{ESS} = [-221, -250, -454, 15, -46, -500]$	Fault on (49,50), and (54,57)	(18,35), (51,108), and (13,52) opened	$\mathbf{\hat{p}}^{ESS} = [-221, -250, 0, 0, 113, -500]$ $\mathbf{\hat{q}}^{ESS} = [38, 39, 0, 0, 62, 121]$	0.021 sec
10	$\mathbf{p}^{ESS} = [-250, 181, 252, 378, 55, 108]$	(49,50) recovered	(18,35), (51,108) closed	$\mathbf{\hat{p}}^{ESS} = [-250, 181, 252, 0, 55, 108]$ $\mathbf{\hat{q}}^{ESS} = [42, 43, 112, 0, 66, 129]$	0.015 sec
13	$\mathbf{p}^{ESS} = [167, 165, 37, 385, 18, 216]$	Fault on (21,23)	(13,18), (18,35) opened	$\mathbf{\hat{p}}^{ESS} = [167, 0, 261, 0, 99, 216]$ $\mathbf{\hat{q}}^{ESS} = [45, 0, 113, 0, 67, 128]$	0.026 sec
14	$\mathbf{p}^{ESS} = [3, 107, -15, 228, 33, -53]$	(54,57) recovered	(13,52), (97,101) closed	$\mathbf{\hat{p}}^{ESS} = [3, 0, -15, 228, 33, -53]$ $\mathbf{\hat{q}}^{ESS} = [41, 0, 105, 73, 63, 119]$	0.029 sec
23	$\mathbf{p}^{ESS} = [-218, -205, -480, -500, 4, -390]$	(21,23) recovered	(13,18), (18,35), and (97,101) opened	$\mathbf{\hat{p}}^{ESS} = [-218, -205, -480, -500, 4, -390]$ $\mathbf{\hat{q}}^{ESS} = [30, 33, 85, 57, 51, 97]$	0.019 sec

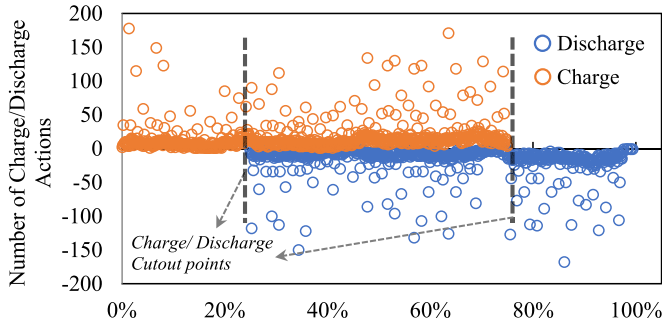


Fig. 7. Number of charging and discharging actions at different ESS energy levels, taken in 10 k generated states. The LC has learned to cut ESS charging/discharging at certain high/low energy levels.

and price patterns in different times of the week. The next important factors are current state of the charge, time of the day, and day of the year.

Another noteworthy observation is how the charging/ discharging actions are taken based on the available ESS energy level. In the 10 k generated instances, the total number of charging and discharging actions at different ESS energy levels are counted and displayed in Fig. 7, which indicates that charging and discharging are stopped at around 75% and 25% energy levels, respectively. Note that no prior knowledge about these cutout levels is embedded in the model, and they are solely learned through trying out different charging strategies during the DRL training process.

In order to evaluate the performance of the hierarchical operation model in the face of sever events, a 24-hour operation scenario is simulated with multiple faults and the control response from both local and grid level controllers is observed. The sequence of failure events, as well as switching responses issued by the CC, preliminary actions made by LCs, and final actions by CC, are shown in Table VI, where the revised actions are shown in bold. In hour 9, two simultaneous faults happen in zones 3 and 4, leading to isolation of these zones as well as zone 5 which is islanded. As a result, ESS in zones 3 and 4 are shut down while ESS5 is set to full discharge within load constraints. The fault on zone 3 is recovered at hour 10, allowing for this zone and zone 5 to be reconnected to the grid. However, zone 4 is still isolated and its ESS is kept off by the CC. The next fault happens in zone 2 at hour 13, which combined with an existing fault on

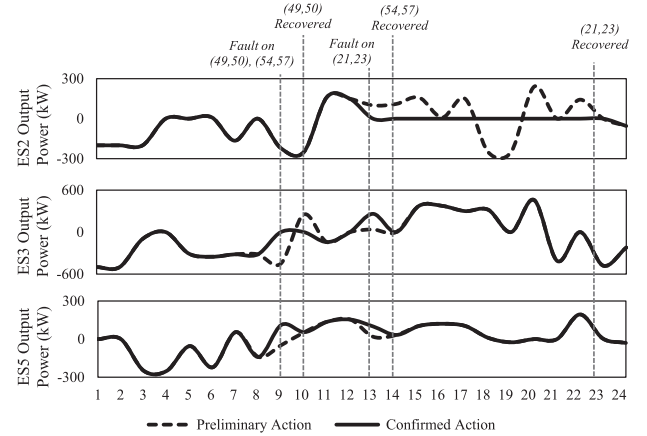


Fig. 8. Actions made by LCs, and confirmed ones by the CC, during the sequence of failure events, on test energy data from Feb. 14, 2018.

zone 4 creates a more widespread outage in zones 2, 3, 4, and 5. While ESS in zones 2 and 5 are shut down, ESS3 and ESS4 are set to full discharge, resulting in partial load restoration in zones 3 and 4. As shown in the Table VI, the total computation time at each step is in the range 0.015 ~ 0.029 seconds which is almost real-time operation. The impact of faults and their recovery on preliminary and confirmed actions of affected ESS is shown in Fig. 8, indicating that CC intervenes to modify preliminary actions only during events that threaten the grid security.

V. CONCLUSION

This paper developed a hierarchical intelligent operation model for scheduling distributed ESS in distribution grids, in which a local controller is trained by DRL on real-world data to learn how to make farsighted decisions on active power dispatch of ESS, while a central controller conducts critical system-wide operations including voltage control and fault management. The proposed hierarchical operation model was tested on a relatively large test system and was found efficient in finding real-time near-optimal decisions for scheduling numerous ESS units, and effective in taking safe critical decisions such as fault isolation. It was shown that the estimated reward and taken actions by DRL-trained local controllers are sensitive to many environmental factors, including day of the week, day of the year,

and predicted price, which highlights the importance of using a data-driven method. DRL-trained local controllers have also established their own charging and discharging cutting points to efficiently schedule the operation of ESS. Further, the TD3 method adopted to implement the DRL for local controllers, outperforms the DDPG method and results in a hierarchical model that also outperforms stochastic programming, in terms of long term operation costs. Also, the computation time of the hierarchical model is at least three orders of magnitude faster than optimization-based methods, making it an ideal alternative for real-time applications. This work opens up directions for the use of similar hierarchical structure for other operational tasks in power systems such as frequency control and dynamic stability as well as hierarchical charging mechanisms for electrical vehicles.

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